

SAI PRANATHI Data & AI Engineer | Software Developer

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PROFESSIONAL EXPERIENCE

Software Development Engineer, Caterpillar  01/2025 – Present

Root Cause Analysis - Python, MCP framework, SQL

- Developed a bug triaging system leveraging MCP framework and Copilot's agentic features, integrating Snowflake and AWS S3 as data sources.
- Automated root cause identification and report creation to optimize debugging workflows and accuracy.

Query Modernization & Optimization – Python, SQL, Agentic AI

- Automated SQL query optimization and refactoring for queries provided in CSV files using instruction-driven mapping documents to migrate to updated table names and field mappings.
- Reduced manual effort and improved consistency in large-scale query remediation by implementing agentic workflows for query transformation and validation.

Quality Assurance Dashboard – SQL, Power BI, Snowflake, Python

- Built a statistical anomaly detection framework to identify row count deviations, attribute pattern mismatches, and behavioral changes in critical data attributes, enabling proactive data quality monitoring and faster issue detection.
- Developed a Power BI dashboard backed by Snowflake to track, visualize, and manage cross-domain test execution results, with automated query execution and result storage using Snowflake stored procedures.

Data and AI Consultant Intern, Thorogood Associates  06/2023 – 11/2023

Property & Spatial Data Insights - Pyspark, ADF, Databricks

- Worked for a top property development firm's analytics team focusing on retail and spatial data insights extraction by utilizing Databricks Notebooks managed by ADF.
- Managed metadata and configuration using SQL to support and optimize data workflows.

Sustainability - Pyspark, PowerBI, ADF

- Collaborated with the Internal Sustainability team to extract workplace carbon emission insights by analyzing key areas such as transport and energy via Databricks Notebooks orchestrated through ADF.
- The processed data was then used in Power BI to create impactful visualizations supporting sustainability initiatives.

EDUCATION

M.Sc (Integrated) Software Systems, PSG College of Technology 
9.10 / 10 CGPA

TECHNICAL SKILLS

Core Competencies

Data Analytics, AI Solution Developer, Data Engineering

Cloud & Data Platforms

AWS, Snowflake, Data Bricks, Azure Data Factory

Frameworks & Tools

MCP, Pyspark, Docker, GitHub, GitHub Copilot

Languages

Python, SQL, C, C++

Visualization

PowerBI, Streamlit

RECOGNITIONS & CERTIFICATIONS

- Outstanding performer, KLA Tencor Hackathon - 2023
- Special Mention team | CAT Early talent Hackathon - 2025
- Generative AI Fundamentals in Databricks - 2023